Elton Chang

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EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

RESEARCH INTERESTS

Contact mechanics, motion prediction and planning, and reproducible robotics experiments.

INDEPENDENT RESEARCH

Tactile Prediction and Contact Mechanics Oct 2026 - Present

- Implemented Python/PyTorch models and retrieval/regression baselines for tactile vibration prediction; ran matched experiments across speeds, directions, loads, and observation budgets.
- Designed surface-level evaluation with wrong-support controls and paired bootstrap intervals; compared eight force-prediction methods and controls on 450 synthetic trials across 10 groups.
- Developed analytical coasting/stopping models and a bounded release-speed planner for target distances, with unit/sign-aware data ingestion and provenance checks.
- Reproduced 1,878 sensor windows and 22 prediction arrays; documented methods, results, and limitations in research reports. Physical validation remains pending.

Squid | Visuo-Tactile Collection Stack Sep 2026

- Implemented C++ RP2040 firmware for an SPI jaw encoder and multiplexed strain sampling, with a binary USB protocol, protocol tests, and error handling.
- Built Python ROS 2 Humble publishers, synchronization monitoring, and MCAP recording for tactile/force streams; added simulated sources, fault injection, and offline data-quality checks.

PROJECTS

Next Best Action (CMU/PNC Collaboration)

- Compared HistGradientBoosting, LightGBM, XGBoost, and ensembles for next-month product adoption; achieved 0.811 ROC AUC and a 95% top-three hit rate on supported recommendations.

Quantifying Meritocracy (Research)

- Extended the agent-based Talent vs. Luck model with a solidarity parameter; ran reproducible simulations to evaluate redistribution using inequality, mobility, and wealth concentration metrics.

TraceFrame

- Developed Python tools for dataset checks, lineage, and HTML audit reports across pandas, Polars, and DuckDB workflows, supporting reproducible analysis and research documentation.

EXPERIENCE

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA

- Improved XGBoost aircraft-on-ground risk ranking (Precision@20: 35% to 62%); tuned model parameters and thresholds with Optuna, increasing median warning lead time from 6.5 to 10.2 days.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA

- Standardized testing environments across 15+ non-production hosts with Terraform; built backup/replication telemetry and alerting, reducing anomaly detection time by 40%.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA

- Automated file ingestion, metadata indexing, and lifecycle tracking in a cloud-hosted file management system using scheduled Python jobs.

TECHNICAL SKILLS

Proficient Tools and Languages: Python, C++, Rust, SQL, PyTorch, scikit-learn, NumPy/SciPy
Robotics: ROS 2, sensor synchronization, contact dynamics, motion prediction, experiment design